

Delegated Emergency Authority as Bounded Executive Action: A Formal Grammar for Time-Limited Constitutional Amendments

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Abstract

Delegated emergency authority has, across two thousand years of constitutional practice, exhibited a consistent failure mode: an authority granted as a time-bounded exception ratchets, through use and reuse, into a permanent feature of the legal order. The pattern recurs in regimes that have very little in common doctrinally. Article 48 of the Weimar Constitution was an emergency-decree authority intended for narrowly delimited use; it was used 250 times in twelve years before being absorbed into the structure of the Nazi state. The Authorization for Use of Military Force enacted in September 2001 has, twenty-five years later, been cited as legal basis for armed action in at least eight countries. Statutory takedown regimes under Section 230 of the Communications Decency Act and Article 17 of the General Data Protection Regulation produce orders that are, in form, time-bounded executive acts; in practice the orders are irreversible by construction because no rollback path is statutorily required. The Foreign Intelligence Surveillance Act's emergency authorities have been documented, in declassified Foreign Intelligence Surveillance Court opinions and Privacy and Civil Liberties Oversight Board reports, to operate at a boundary of judicial review that the original statute did not contemplate.

This Article argues that the ratcheting pattern has a structural rather than merely political explanation. Each of these regimes shares a missing element at the level of statutory grammar: the authority is granted without a typed rollback obligation that must be constructed at the moment of authorization. A typed rollback witness is a formal-methods construct, drawn from recent work on machine-checkable governance receipts. It is the requirement that an authority cannot be exercised unless, at the moment of its exercise, the actor exhibits a proof that the path back to the prior state is constructible. The grammar makes the corresponding class of unbounded emergency authority *unconstructible* as a matter of construction rather than only forbidden. The distinction matters because forbidden authority can be exercised against the prohibition; unconstructible authority cannot be exercised because the construction itself fails.

The Article applies the grammar to five case studies, situates it within the Schmitt-Agamben tradition on the state of exception, and exhibits one implementation drawn from a Lean-attestable governance substrate. The implementation is given as proof that the grammar is buildable, not as the centerpiece of the argument. The Article is structural; it does not defend any particular emergency authority on the merits. It argues that, where the structural grammar is available, its absence in current statutory practice is a defect for which the formal substrate provides a corrective. The Article's claim therefore is not abolition or expansion of any particular emergency authority but reformulation: any future grant should be enacted with the four-component grammar this Article specifies.

*Draft circulating for legal-academy review. The substrate cited in Section 5 is developed at length in a companion technical paper. Comments on the legal claims, particularly the case-study selection in Section 3, are welcomed at the address provided to the editors.

I. Introduction

A familiar shape recurs across the history of delegated emergency authority. A legislature, faced with a circumstance it characterizes as exceptional, grants an executive officer a power that was not previously available within the ordinary structure of the legal order. The grant is framed as time-bounded. The authority is to be used only while the exceptional circumstance persists, and is to lapse, by its own terms, once the circumstance abates. The grant is sometimes accompanied by a sunset provision, sometimes by a periodic reauthorization requirement, sometimes by reporting obligations to a legislative committee, and sometimes by judicial review of the executive's exercise of the granted authority. In each case, the structural intuition motivating the grant is that the authority is borrowed, not given outright; that it must return to the legislature when the exception ends; and that the legal order will resume its ordinary shape thereafter.

The structural intuition is, with regularity that does not appear to be coincidental, falsified. Article 48 of the Weimar Constitution authorized the Reich President to take “the necessary measures” to restore public security in cases where it was “substantially disturbed or endangered” [?]. The article required notification to the Reichstag and permitted the Reichstag to demand the suspension of any measure taken under it. In its first twelve years of operation, the authority was invoked approximately 250 times [?]. By the time of the Reichstag Fire Decree in February 1933, the authority had become a routine instrument of governance rather than the exceptional remedy the constitutional text described [?]. The transition from emergency power to instrument of dictatorship occurred through a sequence of contested political decisions, each conducted within a constitutional grammar that did not require lapse-by-construction. There was no single moment at which the authority's character changed. There was, instead, a sequence of admissions, each of which appeared, at the time of its admission, to be a defensible exercise of a power constitutionally available to its holder.

The Authorization for Use of Military Force enacted on September 18, 2001 [?] permits the President to use “all necessary and appropriate force” against those determined to have authored or aided the September 11 attacks. The text contains no sunset. It contains no geographic limit. It is, twenty-five years later, the primary legal authority for armed action by United States forces in countries that did not exist as targets of policy when the authorization was enacted [?, ?]. The authorization has been cited as legal basis in at least eight named theaters by 2024 [?]. Periodic legislative efforts to repeal and replace the 2001 AUMF have not produced enacted legislation. The pattern admits both a political and a structural reading. A run of bipartisan repeal-and-replace proposals, including Kaine-Young, Lee-Murphy, and the 2023 repeal of the 1991 and 2002 Iraq AUMFs that left the 2001 AUMF intact, demonstrates that the political coalition for repeal has at times assembled [?, ?]. What has not converged is agreement on a successor authority. The structural reading this Article advances does not displace the political account; it identifies an additional feature of the original grant that lowers the cost of continuation relative to replacement at each successive period.

Section 702 of FISA, codified at 50 U.S.C. § 1881a, governs programmatic acquisition of foreign-intelligence information from non-US persons reasonably believed to be located abroad. The program is not itself an emergency authority, but § 1881a(c)(2) contains a sub-provision permitting the Attorney General and Director of National Intelligence to authorize targeting prior to FISC approval of a new certification. That sub-provision, and its interaction with the broader § 702 retention and querying framework documented in

declassified FISC opinions and PCLOB reports [?, ?], exhibits the structural pattern this Article identifies. The structural defect is a grant of authority intended for time-bounded use that contains no mechanism by which the underlying surveillance is rolled back when the emergency lapses. The information collected during the emergency window remains in the holdings of the collecting agency. The targets do not return to a prior state.

Article 17 of the General Data Protection Regulation grants data subjects a right to erasure of their personal data in specified circumstances [?]. The Court of Justice of the European Union's enforcement of the article has produced erasure orders against search-engine providers and platform operators that, in form, are time-bounded executive acts of administrative agencies. In practice, an erasure order does not contemplate a path back to the prior state. A search-engine index from which a result has been removed is not, on the regulation's default operation, restored, although the underlying balance is re-evaluable and the controller may re-index where the supporting facts change [?]. The order is structurally irreversible at the moment of its admission, even though no statutory text characterizes it as such.

Section 230 of the Communications Decency Act is, in its primary form, a liability shield rather than a takedown statute: § 230(c)(1) immunizes interactive computer services from being treated as the publisher of third-party content, and § 230(c)(2) protects good-faith restriction of access to objectionable material [?]. The formal notice-and-takedown regime in United States law is the Digital Millennium Copyright Act's § 512 framework, which includes a § 512(g) counter-notice procedure that obligates the service provider to restore content within a statutorily specified window absent the filing of suit by the original complainant [?, ?]. Section 230's incentive shadow produces takedown-like behavior in platform self-regulation, though the statute itself does not authorize takedown. The structural shape of the time-bounded executive act, on this Article's reading, locates within the DMCA § 512 regime and within the broader pattern of platform self-regulation that Section 230 shelters [?]. That characterization is one this Article advances against the consensus of the Section 230 academy (Goldman, Citron, Keller), which has, in differing directions, resisted treating Section 230 as a takedown regime [?, ?]. The removed content, once removed, is not statutorily required to be restored upon any later event, and the § 512(g) counter-notice path, although a partial rollback mechanism, is in empirical practice underutilized.

These five regimes do not share doctrinal substance. The constitutional emergency power of a Weimar-era Reich President bears no doctrinal relation to the content-moderation practices of a contemporary American social-media platform, and the surveillance authorities of FISA Section 702 bear no doctrinal relation to either. They share, however, a structural feature that this Article identifies and names: the original grant of authority was made without a typed rollback witness. The authority could be admitted without exhibiting, at the moment of admission, a constructible path back to the state of affairs that existed before the authority was exercised.

The Article's central claim is that this missing structural element is not a peripheral defect. It is the explanation for the ratcheting failure mode that has, in each of these regimes and others, transformed a time-bounded emergency authority into a permanent feature of the legal order. The claim is not that a typed rollback witness, had it been required at the time these authorities were enacted, would have prevented every adverse exercise of the authorities. It is the narrower and more defensible claim that the absence of the typed

rollback witness made the ratcheting pattern structurally available, and that its presence would have made the same pattern structurally unavailable, although the political pressure that produced the underlying decrees would have remained.

The grammar that supplies the typed rollback witness is the grammar of bounded executive action, developed in a companion technical paper that constructs a Lean-attestable substrate for receipt-bounded polities [?]. The companion paper's contribution is the formal substrate. This Article's contribution is the application of the substrate's grammar to the doctrinal pattern of delegated emergency authority, the demonstration that the absence of the grammar is the structural explanation for the ratcheting pattern, and the exhibition of one implementation drawn from the substrate.

Part 2 develops the historical pattern in detail and situates it within the Schmitt-Agamben tradition on the state of exception. Part 3 presents five case studies, with explicit acknowledgement that the case studies vary in how cleanly they fit the framing. Part 4 states the formal grammar and explains the role of the typed rollback witness. Part 5 cites the Lean substrate without re-deriving it. Part 6 exhibits one implementation as a working example. Part 7 acknowledges the limits of the formal approach, particularly the irreducibly political content of the exception that the wrapper cannot capture. Part 8 concludes.

II. The Historical Pattern of Ratcheting Emergency Authority

A. The classical diagnosis: emergency as constitutive of order, or as parasitic on it

Carl Schmitt's *Political Theology*, first published in 1922 in the early years of the Weimar Republic, opens with the assertion that "sovereign is he who decides on the exception" [?]. The sentence is the canonical statement of a tradition that takes the state of exception to be constitutive of sovereignty rather than parasitic on it. On Schmitt's account, the legal order is not capable, by its own resources, of specifying when its ordinary operation must yield to the extraordinary. The decision is, in his terminology, a decision *about* the legal order, not a decision *within* it. The sovereign is the entity that can make this decision, and the entity whose making of it is recognized as authoritative by the political community over which the legal order is asserted.

The Schmittian framing produces a particular relationship between the formal text of an emergency authority and its actual exercise. The formal text, on this account, cannot capture the exception. The text can describe categories of circumstance, can specify procedures, can list the officials authorized to act, and can require notification, reporting, or judicial review. What the text cannot do is decide, in advance and from within the legal order, whether the exception obtains. That decision is constitutive of the order and therefore prior to it.

Giorgio Agamben's *State of Exception*, published in Italian in 2003 and in English in 2005, takes the Schmittian diagnosis and develops it into a claim about the trajectory of twentieth- and twenty-first-century constitutional practice [?]. Agamben's argument is that the state of exception, far from being the rare and bounded condition that classical constitutional theory contemplated, has become the dominant paradigm of governance. The temporary suspension of ordinary law has, over a sequence of historical events, become the permanent operating condition of the legal order. The exception, on Agamben's reading, no longer returns the

order to its ordinary shape after the exceptional circumstance abates. The exception persists, and the ordinary shape recedes.

The Schmitt-Agamben framing is the analytic background against which this Article writes. The Article does not take a position on whether the political-theological reading of sovereignty is correct. It takes Agamben's empirical claim that the exception has, in practice, ceased to return the order to its ordinary shape, and asks whether the persistence has a structural rather than a merely political explanation. For Schmitt, the constitutive character of the decision is normatively defensible under conditions of genuine emergency, not merely descriptive of how legal orders are seen to fail. The structural argument this Article advances is therefore incompatible with that normative view, not orthogonal to it: a Schmittian who accepts the constitutive sovereign decision will read the typed rollback witness, and the disciplined wrapper around emergency authority more generally, as the Kelsenian-formalist disease that the 1922 essay diagnosed, not as a corrective [?, ?].¹

B. The structural reading of the ratcheting pattern

The historiography of the Article 48 trajectory remains contested. Kershaw, Mommsen, and Caldwell differ on whether the 1930-32 emergency-rule period under Brüning was structurally entailed by Article 48's text or politically contingent on Hindenburg's preferences and SPD tactical toleration [?, ?, ?]. This Article advances the structural reading without claiming that the structural absence forces the political outcome. The weaker claim is that the absence made the outcome structurally available, not that it made the outcome inevitable. The converse Schmittian position canvassed in the preceding subsection, on which decisionism is a normatively defensible response to genuine emergency rather than a pathology, remains available to a reader who rejects the structural reading on substantive grounds.

Once the question is posed in structural terms, the historical record yields a recurring shape that does not appear to be reducible to political contingency alone.

The shape is the following. A grant of authority is enacted at moment t_0 , on the legislative understanding that the authority will be exercised during a circumstance c that obtains at t_0 , and that the authority will lapse at moment t_1 when c no longer obtains. The grant of authority does not, however, include a structural mechanism that makes the lapse automatic. The lapse is a political event, requiring affirmative action by some body to either refrain from extending the authority or to enact its termination. The continuation of the authority is the default; the lapse is the deviation from the default.

The shape produces a particular kind of equilibrium failure. At each moment $t_i > t_0$, the political question facing the body authorized to extend or terminate the authority is the question of whether to disturb the present operating arrangement in which the authority is active. The disturbance carries political costs, and the maintenance of the present arrangement does not. The expected political cost calculation favors maintenance. Each individual maintenance decision is, in isolation, a defensible exercise of legislative discretion. The cumulative effect of a sequence of such decisions is the transformation of an authority that

¹The Article is not unsympathetic to the structural reading of the political-theological tradition, on which see also Kahn's account of political theology in liberal constitutionalism [?]. What follows, however, is an argument about formal grammar, which does not require resolution of the underlying jurisprudential dispute.

was intended as exceptional into an authority that is permanent.

Bruce Ackerman's account of the constitution of emergency, particularly in his post-September 11 writing [?], identifies this equilibrium failure and proposes a structural correction: a constitutional emergency power should be enacted with a built-in supermajority requirement for extension, such that the default at each successive period is termination rather than continuation. Eric Posner and Adrian Vermeule, in *Terror in the Balance* [?], argue that the Ackerman correction would impose costs that are not justified by the empirical record of post-emergency governance. Cass Sunstein's writing on sunset clauses [?] occupies a position closer to Ackerman than to Posner-Vermeule, holding that structural reversion to the default of termination is a desirable feature of an emergency-power regime.

The Article takes the Ackerman-Sunstein side of this debate as background, but adds a feature that the existing literature on sunset clauses and termination defaults does not directly address. The feature is the requirement that the authority, at the moment of its exercise, exhibit a constructible path back to the state of affairs that existed before the exercise. A sunset clause requires the authority to lapse at a specified time. It does not require that the actions taken under the authority be reversible.

The distinction is consequential. The information collected under a surveillance authority does not return to a pre-collection state when the authority lapses. The content removed from a platform index under an erasure authority does not reappear in the index when the order lapses. The territory occupied under an authorization for use of military force is not restored to its pre-occupation state when the authorization lapses, if it ever does. The authorities are time-bounded in their period of grant. They are not time-bounded in their effect.

The structural correction this Article proposes is not a sunset clause. It is a typed rollback witness: a requirement that the authority cannot be exercised unless, at the moment of exercise, the actor exhibits a proof that the path back to the prior state is constructible. The discipline applies at the moment of exercise rather than at the moment of lapse. An authority that, at the moment of exercise, cannot exhibit a path back to the prior state fails to type-check. The exercise itself becomes unconstructible.

The structural correction is, in form, the same discipline that the substrate cited in Part 5 applies to the admission of a receipt into a polity's history. The substrate requires, at the moment of admission, the construction of a proof that the receipt preserves the backward-refinement property over the polity's prior history. A receipt that does not exhibit the proof cannot be admitted. The grammar applies at the admission step rather than at the audit step.

The remainder of the Article develops the implications of this structural correction for delegated emergency authority, exhibits its operation against five case studies, and addresses the principal objections. The objections are real and the Article acknowledges them. The most serious objection is that the political content of the exception is not reducible to a typed construction, and that the formal grammar therefore misdescribes what is happening when an emergency authority is exercised. The Article's response is that the grammar describes the wrapper, not the content, and that even on the Schmittian view that the content is irreducibly political, the wrapper has structure that the substrate captures.²

²The wrapper-content distinction does not resolve the underlying dispute about whether the content of the exception is itself a matter of decision rather than law. It allows the Article to make a narrower structural claim that is compatible with either side of that dispute.

III. Five Case Studies

Each case study below is organized around three questions. First, was the authority, on its statutory or constitutional face, intended as time-bounded? Second, did the ratcheting pattern in fact occur, in the sense that an authority granted as exceptional became routine or permanent? Third, would the typed-rollback grammar developed in Part 4 have structurally prevented the ratchet? The third question is the most demanding and the one on which the Article's structural claim must stand or fall. The Article is candid where the answer is contested or where the case study fits the framing less cleanly. The five case studies vary in fit, and the variation is itself instructive.

A. Weimar Article 48: the canonical case

Article 48 of the Weimar Constitution [?] authorized the Reich President to take “the necessary measures” to restore public security and order in cases where it had been “substantially disturbed or endangered.” The article required the Reich President to notify the Reichstag of measures taken, and Article 48(3) permitted the Reichstag to demand the suspension of any such measure. That suspension power, however, was politically conditional rather than a static institutional check: it required a parliamentary majority willing to assemble against the sitting government, and across 1930 to 1933 the SPD's tactical *Tolerierung* of Brüning withheld that majority, while Hindenburg's dissolutions of the Reichstag in July 1930 and again in 1932 removed the occasion for a suspension vote at precisely the moments it would otherwise have been most consequential. The drafters of the article, working within the constitutional convention at Weimar in 1919, appear to have understood the authority as a narrowly bounded emergency power designed to provide the new republic with a tool for response to acute civil disorder [?, ?, ?].

The historical trajectory of the article's use is well-documented. Through the 1920s, the authority was invoked in response to discrete crises: the hyperinflation of 1923, the political assassinations of the early decade, and the recurring municipal-level civil disturbances that characterized the era. The invocations during the early period were generally followed by Reichstag review, and the measures taken were, in many cases, subsequently regularized through ordinary legislation or allowed to lapse [?].

The pattern shifted in the early 1930s, and the period 1930 to 1933 is best disaggregated rather than treated as a single phase. Heinrich Brüning's government (March 1930 to May 1932) pursued a legalistic, deflationary program of emergency budgets enacted by Article 48 *Notverordnung* after the Reichstag could not assemble a majority for ordinary austerity legislation [?]; the government relied on Social Democratic *Tolerierung* – a tactical decision by the SPD parliamentary group not to invoke Article 48(3) suspension – and Hindenburg's dissolution of the Reichstag in July 1930 removed the immediate occasion for a suspension vote. Franz von Papen's government (June to December 1932) took an openly anti-parliamentary turn, including the July 1932 *Preußenschlag*, the Article 48-based deposition of the elected Prussian government that became the subject of *Preußen contra Reich*. Kurt von Schleicher's brief chancellorship (December 1932 to January 1933) attempted a *Querfront* coalition strategy that collapsed without producing comparable Article 48 measures. Across the three governments the article's text – which spoke of measures necessary to restore public order – had been stretched to cover budgetary decisions, currency policy, and the regulation of

commerce [?, ?]. The Reichstag Fire Decree of February 28, 1933 [?], issued under Article 48 in response to the burning of the Reichstag building, suspended civil liberties indefinitely and authorized the *Schutzhaft* that produced the immediate arrest of approximately eighty-one Communist deputies and the harassment of Social Democratic deputies. The Enabling Act of March 24, 1933 [?] was procedurally distinct, but its two-thirds threshold was reached because Article 48 had already been used to physically remove the KPD opposition from the chamber; the path from Article 48 to the Enabling Act ran through the Reichstag Fire Decree's preventive-detention provisions rather than through any generalized normalization of emergency rule.

The historiography on whether the Article 48 ratchet was structural or political is contested. Ian Kershaw [?] treats the trajectory as substantially political, emphasizing the role of specific political actors and the contingency of particular decisions. Hans Mommsen [?] treats it as more structural, emphasizing the way that the article's text and the constitutional design surrounding it provided no built-in mechanism for the lapse of decreed measures. Peter Caldwell's work on popular sovereignty in Weimar [?] occupies an intermediate position, and the primary-source controversy is collected in Jacobson and Schlink's anthology of Weimar legal-theoretical writing [?]; David Dyzenhaus reconstructs the Kelsen-Schmitt-Heller debate that runs alongside the constitutional record [?]. The Article does not need to resolve this dispute. It needs only the narrower observation that Article 48 contained no typed rollback obligation: the Reich President could exercise the authority without exhibiting, at the moment of exercise, a constructible path back to the prior state. The decrees taken under the authority created facts on the ground – suspended civil liberties, imprisoned dissidents, dissolved political organizations – that did not return to the prior state when the formal emergency ended.

The constitutional order was not, however, wholly passive in the face of the trajectory. *Preußen contra Reich*, decided by the Staatsgerichtshof on 25 October 1932 [?], was the case in which Article 48's limits were actually litigated. The Land of Prussia challenged the Papen government's July 1932 *Preußenschlag*, the Article 48-based deposition of the elected SPD-Centre Prussian government and its replacement by a Reich Commissioner. The court partially upheld the Reich's authority to install a commissioner for limited functions but partially struck down the deposition, holding that the Prussian ministers retained their constitutional status as representatives of the Land in the Reichsrat and that the Reich could not, under Article 48, displace them from that representative capacity. The decision is direct evidence that the constitutional order resisted the ratchet in a way the purely structural narrative elides: the limits of Article 48 were not only textually unspecified but also judicially contestable, and the contest produced a partial substantive ruling against the executive within months of the action. The structural reading this Article advances must accommodate that ruling. The accommodation is that the structural absence of a typed rollback witness made the ratchet structurally available, in the sense that nothing in the text or surrounding architecture made the ratchet unconstructible; *Preußen contra Reich* demonstrates that the constitutional order's resistance to the ratchet was politically contested and partially successful, which is consistent with the weaker structural reading but inconsistent with a strong claim that the ratchet was textually entailed.

The typed-rollback grammar would have structurally constrained the article's use in a way the original text did not. A grammar that required the Reich President, at the moment of exercising Article 48 authority,

to exhibit a witness for the path back to the prior state would have made decrees of that form unconstructible, although the political pressure that produced the decrees would have remained. The grammar would not have prevented the political use of the article to circumvent the Reichstag, but it would have prevented the conversion of that political use into permanent constitutional facts. This is the cleanest application of the framing to a historical case, and the case study where the structural absence is most legible.

B. Section 230's incentive shadow and the DMCA 512 takedown regime

The framing of Section 230 of the Communications Decency Act [?] as bearing structural similarity to a delegated emergency authority is one this Article advances against the consensus of the Section 230 academy, which has, in differing directions, resisted the characterization of 230 as a takedown regime [?, ?, ?]. Eric Goldman has argued that 230 is best understood as a non-takedown immunity that decouples platform incentives from content-moderation outcomes [?]; Danielle Citron and Benjamin Wittes have argued for narrowing the shield on bad-Samaritan grounds rather than on irreversibility grounds [?]; Daphne Keller's empirical work on over-removal under intermediary-liability regimes argues against treating immunity withdrawal as a takedown lever [?]. The Article's structural claim is therefore advanced with the academy's resistance in view, and the characterization that follows separates the components of the regime more carefully than its prior formulations did.

Section 230(c)(1) is a liability shield: it operates *ex post*, when a plaintiff sues, and immunizes a publisher from being treated as the speaker of content posted by another. It does not authorize, require, or even mention takedown. Section 230(c)(2) is structurally distinct from (c)(1). The "good Samaritan" carve-out in (c)(2) explicitly immunizes good-faith action to restrict access to material the provider considers obscene, lewd, lascivious, filthy, excessively violent, harassing, or otherwise objectionable. That sub-section is, on its face, a hedged authorization of takedown by the private intermediary, paired with no statutory restoration obligation. The Article treats (c)(2) honestly: it is already a takedown regime of a particular form, and the absence of a structural rollback obligation within (c)(2) is one element of the pattern this Article identifies.

The Digital Millennium Copyright Act's notice-and-takedown procedure [?] is the operative notice-and-takedown regime in U.S. intermediary-liability law, and unlike Section 230 it contains a structurally relevant partial rollback path: 17 U.S.C. § 512(g) obligates the service provider to restore content within ten to fourteen business days of a properly formed counter-notice unless the complainant files suit [?]. The § 512(g) counter-notice is, in the formal sense this Article develops, closer to a typed rollback witness than either Section 230(c)(2) or the platform-self-regulatory takedowns that occur in the broader shadow of immunity supplies; it is a procedurally required path back to the prior state, even if the path is conditional on the user's willingness to incur the friction of filing. The Article therefore does not claim that the intermediary-liability landscape lacks any rollback-adjacent apparatus. It claims, more narrowly, that Section 230's incentive shadow – the (c)(1) immunity coupled with the (c)(2) good-faith carve-out – produces platform self-regulatory takedowns that operate outside the DMCA's structured notice-and-counter-notice framework, and that those self-regulatory takedowns lack the structural rollback witness that § 512(g) supplies in the copyright-takedown adjacent regime.

The empirical record on takedown patterns is consistent with this narrower characterization. Work on

DMCA notice-and-takedown indicates that a substantial fraction of notices target content for which the underlying infringement claim is dubious or absent [?], and that the § 512(g) counter-notice is rarely used because of the friction it imposes on the user whose content was removed [?]. The counter-notice regime is, in other words, a typed rollback witness with high enough activation costs that the underlying default approximates non-restoration; the platform's safe-harbor incentive favors removal, and the counter-notice path is available but underutilized. In the broader self-regulatory regime that operates in Section 230's incentive shadow, no analogous witness exists at all: removals based on internal policy rather than statutory notice are not subject to a § 512(g)-style restoration path.

The typed-rollback grammar, applied to this regime, identifies the structural locus precisely. The grammar does not displace § 512(g) in the DMCA-counterpart context; rather, it generalizes § 512(g)'s logic and presses it back to the moment of removal. A removal action under either the DMCA or the broader Section 230 self-regulatory shadow would, under the grammar, require the platform to construct, at the moment of removal, a witness for restoration of the content in the case that the underlying claim is later determined to be without merit. The witness would have to be activated structurally rather than only on user objection. A removal that cannot be reversed by construction would fail to type-check.

The application of the grammar to this regime is contested. Goldman would observe that the grammar imposes a structural cost on intermediaries that Section 230 deliberately declines to impose; Keller would observe that the typed-rollback discipline does not address the over-removal pattern her empirical work documents because it operates at the moment of admission rather than at the moment of incentive formation; Citron and Wittes would argue, from a different direction, that the discipline does not address the harms produced by content that should have been removed but was not. The Article acknowledges these objections without resolving them. The structural claim is the narrower one: that Section 230's incentive shadow produces platform self-regulatory takedowns that lack a structural rollback witness, even though DMCA § 512(g) provides a partial counter-example in the adjacent copyright-takedown regime. The policy question of whether the grammar should apply to this regime is a separate question on which the Article does not take a position.

C. GDPR Article 17: erasure without rollback

Article 17 of the General Data Protection Regulation [?] grants data subjects a right to erasure of their personal data in specified circumstances. The article's enforcement has produced a body of erasure and de-referencing orders against search-engine providers and platform operators, of which Case C-131/12, *Google Spain v. AEPD*, ECLI:EU:C:2014:317 (13 May 2014) [?] is the foundational example. The erasure orders are time-bounded executive acts in the formal sense that they are issued by administrative bodies operating under defined procedures, and they are subject to judicial review on familiar administrative-law grounds. The Court of Justice has, since *Google Spain*, developed a substantial body of jurisprudence operationalizing the balance the controller and the data-protection authority must strike. Case C-136/17, *GC and Others v. CNIL*, ECLI:EU:C:2019:773 (24 September 2019) [?] clarified the scope of de-referencing for special-category data and held that the balance between data-subject rights and public-interest access is re-evaluable in light of changed circumstances. Case C-507/17, *Google LLC v. CNIL*, ECLI:EU:C:2019:772 (24 September 2019)

[?] held that the right to de-referencing does not, as a matter of EU law, extend to all domain extensions globally. Case C-460/20, *TU and RE v. Google LLC*, ECLI:EU:C:2022:962 (8 December 2022) [?] addressed re-listing where allegedly inaccurate content is subsequently shown to be accurate, treating the balance as one the controller must reassess. The prior-regime precedent that established the threshold definition of personal-data processing that the GDPR inherited is Case C-101/01, *Lindqvist*, ECLI:EU:C:2003:596 (6 November 2003) [?].

The ratcheting pattern in this regime takes a different form from the Weimar case. There is no observed trajectory in which the erasure authority has been used to dissolve the underlying constitutional order or to substitute for ordinary legislation. The pattern that this Article identifies is the structural irreversibility of individual erasure orders in the regulation's default operation. An erasure order requires the controller to delete or de-index the targeted data. The deletion is not, on the regulation's default operation, restored, although the underlying balance is re-evaluable and the controller may re-index where the supporting facts change [?]. The order is, in form, a time-bounded executive act. It is, in effect, a permanent alteration of the data ecosystem absent affirmative reassessment by the controller, who is under no structural obligation to re-evaluate at any defined interval.

The Article's framing here is narrower than in the Weimar case. The claim is not that the GDPR Article 17 regime has ratcheted into something other than what it was designed to be. The claim is that individual erasure orders share the structural defect of an emergency authority granted without a typed rollback witness: the order can be admitted without exhibiting, at the moment of admission, a constructible path back to the prior state. The CJEU's recognition that the underlying balance is re-evaluable in cases like *GC and Others* and *TU and RE* establishes the doctrinal possibility of restoration in specific factual configurations; it does not establish a structural witness for restoration at the moment of admission. The witness is not currently a requirement of the regime.

The European Data Protection Board's Guidelines 5/2019 on the criteria of the Right to be Forgotten in the search-engines cases [?] codify the controller's ongoing obligation to revisit the balance, and they supersede the Article 29 Working Party Guidelines on Implementation of the Court of Justice of the European Union judgment on "Google Spain and Inc. v. Agencia Española de Protección de Datos and Mario Costeja González" (WP 225, 26 November 2014) [?]. The Article's claim that the regime is structurally irreversible at the moment of admission must be read against these documents: the regime contemplates reassessment, but the reassessment is not a typed rollback witness in the sense Part 4 develops.

The data-protection academic literature has framed the right structurally rather than transactionally. Orla Lynskey's *The Foundations of EU Data Protection Law* (Oxford 2015) [?], particularly Chapters 6 and 7, treats the right to be forgotten as a structural element of the data-protection regime rather than as a remedy attached to particular harms; Lee A. Bygrave's *Data Privacy Law: An International Perspective* (Oxford, 2d ed. 2014) [?] situates the EU architecture in its comparative-law context. Mantelero's writing on GDPR enforcement [?] and Kuner's analysis of the right to be forgotten [?] both observe the structural irreversibility of individual erasure orders in the regulation's default operation. None of these authors proposes a typed-rollback discipline as the corrective. The Article's contribution to this literature is the identification of the structural similarity to the emergency-authority pattern, and the proposal that the typed-rollback grammar

provides a more principled corrective than ad hoc restoration mechanisms.

The fit of the case study to the framing is intermediate. The GDPR Article 17 regime is not on its face an emergency-authority regime, and the data-protection scholars who have written on it have not, in general, framed it as such. The structural similarity is at the level of the form of authority rather than the substance of the regime. The Article includes the case study because the structural similarity is real and because the corrective the grammar proposes is applicable to the regime even though the regime is not, in its self-understanding, an emergency-authority regime.

D. FISA Section 702: programmatic acquisition and its emergency sub-provision

Section 702 of the Foreign Intelligence Surveillance Act, codified at 50 U.S.C. § 1881a [?], is a programmatic certification regime for the acquisition of foreign-intelligence information from non-U.S. persons reasonably believed to be located outside the United States; the program is reviewed annually by the Foreign Intelligence Surveillance Court under 50 U.S.C. § 1881a(j). Section 702 is not, in itself, an emergency authority. The seventy-two-hour mechanism in 50 U.S.C. § 1881a(c)(2) is one sub-provision, permitting the Attorney General and Director of National Intelligence to determine targeting prior to FISC approval of a new certification. The case study this Article advances is about that sub-provision's interaction with the broader § 702 retention and querying framework, as documented in declassified FISC opinions and PCLOB reports, rather than about Section 702 as a whole being an emergency authority. Reading the case study otherwise would conflate the programmatic certification regime with the narrower within-section emergency mechanism, and the Article disclaims that conflation at the outset.

The operative public record on the program runs through a sequence of declassified FISC opinions and PCLOB reports. *In re DNI/AG 702(g) Certifications*, FISC Mem. Op. (Bates, J., Oct. 3, 2011), declassified August 2013 [?], was the upstream-collection multiple-communications-transaction opinion that found a Fourth Amendment violation in NSA's then-current upstream practices and required minimization changes. *Memorandum Opinion and Order*, FISC (Collyer, J., April 26, 2017) [?] disclosed compliance failures in upstream "about" collection and led NSA to halt about-collection. *Memorandum Opinion and Order*, FISC (Boasberg, J., Oct. 18, 2018), declassified 2019 [?] addressed querying-procedure compliance. *Memorandum Opinion and Order*, FISC (Contreras, J., April 21, 2022), declassified May 2023 [?] addressed FBI query compliance and was extensively cited in the 2023–24 reauthorization debate. The Privacy and Civil Liberties Oversight Board's three reports across the period are the PCLOB *Report on the Surveillance Program Operated Pursuant to Section 702 of the Foreign Intelligence Surveillance Act* (July 2, 2014) [?], the *Recommendations Assessment Report* (January 2017) [?], and the *Report on the Section 702 Surveillance Program* (September 2023) [?]. The reauthorization legislation that re-entered the structural picture in this period comprises the FISA Amendments Reauthorization Act of 2017, Pub. L. No. 115-118 (signed January 2018) [?], and the Reforming Intelligence and Securing America Act, Pub. L. No. 118-49 (April 2024) [?], the latter of which tightened FBI querying procedures and audit requirements.

The structural defect, on the framing this Article develops, is the absence of a typed rollback witness for information collection at the admission step. The seventy-two-hour mechanism authorizes the collection; it does not require, at the moment of authorization, the construction of a path back to the pre-collection state.

The targets of the collection do not return to a pre-collection state when the seventy-two-hour authorization is later reviewed or when the underlying certification is modified. The collected information persists in agency holdings.

Existing law contains rollback-adjacent mechanisms: FISC-ordered destruction of unlawfully acquired material under 50 U.S.C. § 1881a(i)(3), the 50 U.S.C. § 1809(a) criminal bar on use or disclosure of unlawfully acquired information, the five-year retention defaults in approved minimization procedures, and the post-2018 querying procedures. These mechanisms are partial: they operate at the audit and lapse steps rather than at the admission step, they do not bind the upstream collection, and they are not constructively reversible witnesses in the sense Part 4 develops. The structural claim this Article advances is not that the regime contains no reversibility apparatus, but that the apparatus it contains is not of the form a typed rollback witness would supply.

The case-law backdrop frames the litigation posture of the program. *Clapper v. Amnesty International USA*, 568 U.S. 398 (2013) [?], established the standing doctrine that has structured almost all § 702 litigation in Article III courts; the speculative-injury framework Clapper announced makes the structural-irreversibility claim non-justiciable in the form in which the Article’s framing might initially suggest it should be litigated. *United States v. Hasbajrami*, 945 F.3d 641 (2d Cir. 2019) [?], is the principal federal-appellate merits decision on § 702 querying of U.S.-person information; the court treated the querying of incidentally collected U.S.-person communications as a distinct Fourth Amendment event from the upstream acquisition, a posture directly relevant to whether the typed-rollback grammar would apply at the querying step, at the acquisition step, or at both.

The Article’s claim here is the most cautious of the five case studies. The public record on Section 702 acquisition and the seventy-two-hour sub-provision is incomplete. The declassified opinions and PCLOB reports provide an authoritative but partial picture of the program’s operation. A national-security-law scholar with classified-clearance access to the full record may have grounds to dispute the characterization above. The structural claim is offered as a working hypothesis on the basis of the available public record, with explicit acknowledgement that the record is incomplete. The typed-rollback grammar’s applicability to surveillance authorities depends on whether information collection can, in principle, be made reversible by construction. The cryptographic literature on revocable encryption [?] suggests that some forms of collection can be made constructively reversible. Whether the construction is operationally feasible in the surveillance context is an empirical question on which the Article does not take a position.

E. The 2001 AUMF: a quarter-century of unbounded authorization

The Authorization for Use of Military Force enacted by Congress on September 18, 2001 [?] permits the President to use “all necessary and appropriate force” against “those nations, organizations, or persons he determines planned, authorized, committed, or aided” the September 11 attacks. The text contains no geographic limit, no sunset, and no requirement of periodic reauthorization.

The trajectory of the AUMF’s use is documented at length by the Brennan Center [?] and by scholarly accounts in Chesney [?]. By 2024, the authorization had been cited as legal basis for armed action in at least eight named theaters, including territories of states not in existence as targets of policy at the time of the

authorization's enactment. The running list is documented in the Department of Defense's 2018 Section 1264 Report on the legal and policy frameworks guiding the use of military force [?], which named seven countries (Afghanistan, Iraq, Syria, Yemen, Somalia, Libya, and the Philippines) and identified Niger and Kenya in earlier transmittals; in the Congressional Research Service's continuing analysis of the 2001 AUMF and its application [?]; and in the periodic War Powers Resolution reports transmitted to Congress under 50 U.S.C. § 1543 [?]. Jameel Jaffer's compilation of the public legal record on drone operations [?] supplies the legal-theory critique of the associated-forces doctrine through which the authority has been extended. The Islamic State of Iraq and the Levant, which did not exist in any form in 2001, has been characterized as a covered organization under the AUMF on the legal theory that it is a successor entity to al-Qaeda in Iraq. The expansion of the authority through the doctrine of associated forces has been documented in Department of Justice white papers, declassified Office of Legal Counsel opinions, and the periodic war powers reports.

The pattern admits both a political and a structural reading, and the structural reading must be advanced carefully because the political record is not one of legislative inertia. A run of bipartisan repeal-and-replace proposals demonstrates that the political coalition for repeal has at times assembled. The Kaine-Young AUMF repeal-and-replace framework was introduced as S.1228 in the 116th Congress and reintroduced through successive sessions [?]; the Lee-Murphy proposals addressed parallel concerns; the Corker-Kaine 2018 framework attempted a successor-authority structure with sunset and reporting requirements [?]; and in 2023 Congress in fact repealed both the 1991 Authorization for Use of Military Force Against Iraq and the 2002 Authorization for Use of Military Force Against Iraq, which President Biden signed, although the 2001 AUMF itself was left intact [?]. What has not converged is agreement on a successor authority for the 2001 AUMF specifically. The structural reading this Article advances does not displace the political account; it identifies an additional feature of the original grant that lowers the cost of continuation relative to replacement at each successive period. The default at each successive period is continuation, and the political coalition required to enact a replacement is harder to assemble than the coalition required to leave the authority in place; that structural asymmetry is the feature the typed-rollback grammar addresses, and it operates alongside, not in displacement of, the political-equilibrium account.

The typed-rollback grammar's application to the AUMF is the most politically charged of the five case studies. The application is structural: the original AUMF was enacted without a typed rollback witness, which is to say, without a requirement that the President exhibit, at the moment of authorizing the use of force, a constructible path back to the pre-authorization state. The grammar would have required the original authorization to specify the conditions under which the use of force was to terminate and the state of affairs to which the post-termination order would return. A military action that could not exhibit, at the moment of authorization, a constructible path back to the prior state would have failed to type-check.

The political content of this case study is not the Article's subject. The Article does not take a position on the merits of any particular use of force under the AUMF or on the desirability of any particular successor authorization, and the structural reading is supplementary to, not a substitute for, the political account that Kaine-Young, Corker-Kaine, and the 2023 Iraq AUMF repeal record represent. The structural claim is the narrower one: that the original AUMF's grant of unbounded authority is a clear instance of the ratcheting pattern, and that the typed-rollback grammar would have made the ratchet structurally less available by

raising the cost of continuation relative to replacement. A reader who disagrees with the structural claim on this case study can disagree with it without disturbing the Article’s claims about the other four cases. The AUMF is included because it is the most prominent contemporary American example of the ratcheting pattern and because a structural account that omitted it would be incomplete; it is also the case study where the political account is strongest, and the Article concedes that the structural reading must ride alongside the political one rather than displace it.

IV. The Formal Grammar: Bounded Executive Action with Typed Rollback Witnesses

A. The construction

The grammar developed in this Part is small. A bounded executive action is a four-tuple

$$a = (\text{act}, \text{TTL}, w, q)$$

where *act* is the action to be performed, *TTL* is the time-to-live bounding the action’s period of effect, *w* is a typed rollback witness, and *q* is the quorum predicate specifying which actors must concur in the action’s admission. The action *a* is admissible into the polity’s history only if the four-tuple is well-formed in the sense defined below. The four components correspond, on the legal side, to the doctrinal categories of substantive action, sunset, reversibility, and authorization procedure. The first three are the components for which the existing statutory record provides systematic instances. The fourth, the quorum predicate, is the component that captures the Ackerman-Sunstein structural-correction proposals on supermajority defaults for emergency-power continuation.

The typed rollback witness *w* is the load-bearing component of the construction. A typed rollback witness for an action *act* is a proof, constructible at the moment of admission of *a*, that the polity’s state after the execution of *act* admits a transformation back to its state before the execution of *act*, and that the inverse transformation is itself an action the polity is authorized to perform. The witness is not a promise to roll back, nor a procedure by which a later actor may request rollback. The witness is a proof that the rollback is constructible.

The distinction between a promise and a proof matters. A promise is a statement about future action that is binding only insofar as the political process that enforces it operates as designed. A proof is a constructive object that, by its existence, demonstrates that the future action is available. A promise can be broken; a proof either exists or does not. An action admitted without a proof of its rollback’s constructibility may turn out, at the moment rollback is required, to be irreversible. An action admitted with a proof of its rollback’s constructibility is, by the construction of the proof, reversible by available means.

The grammar makes the unconstructive case unconstructible. An action *a* for which no rollback witness *w* exists cannot be constructed as a bounded executive action. The exercise of an authority granting such an action fails to type-check. The failure is not at the audit step, after the action has been taken, but at the admission step, before the action has been taken. The typed-rollback discipline applies at construction time, not at execution time.

B. The discipline applied to the case studies

Each of the five case studies in Part 3 is, on the grammar's framing, a case in which one or more components of the four-tuple is absent from the statutory record.

Weimar Article 48 supplied the substantive component (act), in the form of the authority to take "necessary measures." The article did not supply a typed rollback witness (w), and the time-to-live (TTL) was implicit and politically determined rather than typed. The quorum predicate (q) was the Reichstag's authority to demand suspension, which was a post-hoc check rather than a precondition of admission. The grammar's reading of the article is that the action component was present but the other three components were either absent or unenforced at the type level.

The DMCA-Section 230 takedown regime supplies act (the removal) and an implicit TTL (the action takes effect immediately and remains in effect until or unless restored). The witness w is not required at construction time. The counter-notice procedure is a post-hoc check, not a precondition of admission. The quorum q is a single private actor, the platform's content-moderation function.

GDPR Article 17 supplies act and a procedural q (the administrative process by which an erasure order is issued). The witness w is, in practice, not constructible at the moment of admission because the erasure is, in the present technical and operational regime, irreversible by available means. The grammar's reading is that the construction of an Article 17 erasure order, on the present technical regime, fails the typed-rollback test by default.

FISA Section 702 supplies act (the surveillance authorization) and an implicit q (Attorney General plus Director of National Intelligence). The witness w is absent because the surveillance, once executed, produces a state of affairs – information in the holdings of the collecting agency – for which no constructive rollback is statutorily required and, on the present technical regime, no constructive rollback is operationally available.

The 2001 AUMF supplies act (use of military force) and a q (presidential determination of the target's covered status). The witness w is absent. The TTL is absent in the statutory text. The grammar's reading of the AUMF is that three of the four components are either absent or absent at the type level, with only the action component present in statutory form.

The grammar's prescription, in each case, is structural rather than substantive. The grammar does not specify what the substantive action should be, what the time-to-live should be, what the rollback witness should look like, or what the quorum should consist of. It specifies only that all four components must be present, constructible, and type-checked at the moment of admission. The substantive content of each component is a matter of democratic political determination, which the grammar deliberately leaves to the legislative process that enacts the underlying authority.

C. What the grammar does and does not claim

The grammar makes a structural claim. It does not make the substantive claim that any particular delegated emergency authority is desirable or undesirable. It does not specify which actions should be permissible under emergency authority, which time-to-live values are appropriate, what content a rollback witness should have, or which actors should be authorized to concur. These are questions of democratic political choice that

the grammar takes as exogenous.

The grammar also does not claim that the construction is operationally costless. Requiring the construction of a typed rollback witness at the moment of admission imposes a structural cost on the actor seeking admission. In some cases the cost will be small – a platform that maintains versioned content storage can construct a rollback witness for a removal as a routine matter. In other cases the cost will be large – a surveillance authority whose technical implementation does not permit constructive reversal of collection may not be able to exhibit a witness without substantial re-engineering of the underlying collection systems. The cost is a real consideration that any application of the grammar must address.

What the grammar claims is the narrower observation that the cost is currently being shifted onto the polity rather than borne by the actor seeking admission. When an emergency authority is exercised without a typed rollback witness, the cost of the resulting irreversibility falls on the polity, not on the actor. The polity bears the cost in the form of permanent alterations to the state of affairs that the authority was intended to alter only temporarily. The grammar shifts the cost from the polity to the actor. The actor must, at admission, either construct the witness or refrain from seeking admission. The shift is, on the Ackerman-Sunstein structural-correction framing, the appropriate distribution of the cost, because the actor seeking admission is the actor in the best position to bear it.

The grammar’s relationship to the Schmittian objection – that the content of the exception is not formalizable – is that the grammar formalizes only the wrapper. The substantive content of the action, the substantive determination of whether an exception obtains, the substantive question of what state of affairs the polity should return to: these are political determinations that the grammar takes as inputs. What the grammar provides is a structural skeleton within which those determinations are made. The skeleton is constructible even if the content is not. A Schmittian critic of the grammar would observe, correctly, that the most consequential element of an emergency authority is the determination that the exception obtains. The Article’s response is that the determination is preserved as a political determination in the grammar; the grammar adds the requirement that the determination be accompanied by a typed witness for the path back. The witness does not prejudge the determination. The witness conditions only the form of the action that follows from the determination.

V. The Lean Substrate

The grammar developed in Part 4 is constructible. A companion technical paper [?] develops the substrate in detail. This Part sketches the substrate’s relevant features without re-deriving them. The objective is to give the legal-academy reader enough familiarity with the formal machinery to assess whether the grammar is in fact buildable in the sense the Article claims, without reproducing the technical content that a separately published paper has already established.

The substrate is a system in which a polity is represented as a triple (T, C, K) , where T is a receipt-admission scope, C is a Merkle-rooted citizenship roster of the actors authorized to seek admission of receipts into the polity’s history, and K is a finite list of predicates that the polity has adopted as its constitution. A receipt is admissible into the polity’s history only if the predicates in K accept it, and the polity’s history is the totality of admitted receipts.

The substrate provides four operations on this triple that are relevant to the grammar: admission, capability attenuation, treaty intersection, and constitutional amendment. Each operation is implemented in Rust and accompanied by Lean 4 proofs of its key properties. The properties most relevant to this Article’s grammar are the following.

A. Admission as decidable predicate evaluation

A receipt r is admissible into a polity (T, C, K) if and only if $r \in T$ and $K(r) = \text{accept}$. The predicate evaluation $K(r)$ is decidable in the sense that the constitution K is a finite list of predicates and each predicate is evaluable as a Boolean function over the receipt’s canonical representation. The substrate’s runtime kernel implements this evaluation as a fail-closed admission hook: a receipt for which any predicate returns reject is denied, and a receipt for which all predicates accept is admitted and signed.

The grammar’s typed-rollback witness corresponds, on the substrate, to a predicate in K that requires the receipt to carry a rollback obligation. A receipt for a bounded executive action is constructed with fields for act, TTL, w , and q , and the predicate in K that gates admission of such receipts requires all four fields to be present and to type-check. A receipt that lacks a witness w , or that carries a witness that does not type-check against the substrate’s witness specification, fails the predicate evaluation and is denied.

B. Capability attenuation and the executive authority

The substrate represents authority through capability tokens that are signed objects carrying rights. A capability for a bounded executive action carries the action’s specification and is signed by the authority that granted it. The substrate’s capability-attenuation operation permits the holder of a capability to derive a more restricted capability from the original, and a Lean theorem [?] establishes that the derived capability cannot grant rights that the original did not.

The relevance to delegated emergency authority is that the structural relationship between a legislature granting an emergency authority to an executive officer is, on the substrate’s framing, an instance of capability delegation with attenuation. The legislature is the original authority; the executive officer holds an attenuated capability whose scope is bounded by the legislature’s grant. The attenuation theorem ensures that the executive officer cannot, by exercise of the delegated authority, acquire rights the legislature did not grant. The grammar’s typed-rollback witness is an attenuation constraint at the capability level: the executive officer’s capability is bounded by the requirement that any action taken under it carry a witness for the path back.

C. Constitutional amendment and backward refinement

The substrate’s amendment operation transforms a polity from (T, C, K) to (T, C, K') where K' is a candidate replacement constitution. The amendment is admissible only if K' is a backward refinement of K : every receipt admitted under K remains admissible under K' . A Lean theorem on the substrate establishes that the backward-refinement check is decidable and that the substrate’s runtime kernel refuses amendment requests that do not exhibit a refinement witness [?].

The relevance to delegated emergency authority is that an emergency-amendment proposal – a proposal to modify the polity’s constitution in response to an exceptional circumstance – is admissible only if it preserves backward refinement. The amendment cannot, by its own operation, retroactively un-admit a receipt that the polity has previously admitted. The grammar’s typed-rollback witness operates at the action level rather than the amendment level, but the structural intuition is the same: admission of a receipt or amendment requires construction of a witness for compatibility with the polity’s prior history.

D. The substrate’s relationship to the legal claim

The substrate is not the Article’s contribution. The substrate is cited as evidence that the grammar developed in Part 4 is buildable. A legal-academy reader does not need to verify the Lean proofs to assess the Article’s structural claim about delegated emergency authority. The reader needs only to accept that the grammar is, in principle, implementable on a substrate of the kind the companion paper describes. The substrate’s existence answers the objection that the grammar is a theoretical construct without operational meaning. It does not answer the objection that the grammar may be inappropriate for the legal domain to which the Article applies it. That objection is the subject of Part 7.

VI. Implementation in Practice

One implementation exhibits the grammar against an operational workload distant from constitutional emergency powers. In an incident-response context, every containment action is constructed as a four-tuple: the substantive action, a time-to-live, a typed rollback witness, and a quorum predicate.³ The constructibility of the witness is an empirical feature of the operational regime, not an a priori feature of the action’s category (a point that carries over to the legal domains of Part 3). The implementation is an existence proof for the grammar’s buildability; it is not an argument that the grammar should apply to the legal regimes the Article examines.

VII. Limits of the Formal Approach

The Article’s structural claim is bounded. This Part identifies the limits of the claim, the objections to which the Article’s argument is most vulnerable, and the responses that are available against those objections. The Part is offered in the spirit of candor about cross-disciplinary work: a CS-flavored argument applied to legal-doctrinal domains should make its limits explicit, because the asymmetry of expertise between the author’s home discipline and the target audience’s home discipline is large and the failure modes of cross-disciplinary work are well-documented.

³The illustrative workload is endpoint containment. A quarantine carries a constructive path back to the file’s prior location and access-control state; a process suspension carries the inverse signal that returns the process to its running state; an egress restriction carries the prior network policy. Quorum tiers scale with severity, requiring a single operator’s authorization for low-severity actions and concurrent device-level and operator-level signatures within a bounded window for higher-severity ones. Each admitted action emits a signed receipt that is auditable in canonical form.

A. The Schmittian objection

The most fundamental objection to the Article's framing is the Schmittian objection that the content of the exception is not formalizable. The exception is, on Schmitt's account, a decision that is constitutive of sovereignty rather than a determination within an existing legal order. A formal grammar that purports to govern the wrapper of an exception while leaving the content as a political input misdescribes what is happening when an emergency authority is exercised. The decision to invoke an emergency authority is, on the Schmittian view, the politically prior act; the formal structure of the authority is downstream of that decision and cannot constrain it.

The Article's response is the wrapper-content distinction developed in Part 4. The grammar formalizes the wrapper. The content of the exception, including the determination that an exception obtains, remains a political input on which the grammar does not prejudge. On the Schmittian view that the content is constitutive of sovereignty, the grammar is consistent with that view because the grammar adds structure only to the wrapper.

The response is partial. A committed Schmittian critic may argue that the wrapper-content distinction is itself a category mistake: the wrapper is part of the content, and an attempt to formalize the wrapper independent of the content is an attempt to constrain the political decision through formal means that are by hypothesis incapable of doing so. The Article does not have a knockdown response to this version of the objection. The Article's position is that, even on the strong Schmittian view, the formal wrapper is a constraint on the exercise of authority that is distinct from a constraint on the decision to exercise it, and that constraints on the exercise are valuable even if constraints on the decision are unavailable. A reader who finds this response unconvincing is, on the Article's framing, holding a view that the Article does not purport to refute.

B. The legal-realist objection

A related but distinct objection comes from the legal-realist tradition. On this view, the formal text of an authority is a poor predictor of how the authority will be exercised in practice. The actors who exercise the authority will, predictably, develop interpretive practices that bend the text to their operational needs. A typed-rollback discipline imposed by statute would be, on the legal-realist view, subject to the same bending. The witness *w* would, in practice, become a pro forma artifact that satisfied the formal requirement without imposing any meaningful constraint on the action it accompanies.

The legal-realist objection is substantial. The empirical record on procedural requirements in administrative law and constitutional law is mixed. Some procedural requirements have been highly effective in constraining the actions of officials (notice and comment in administrative rulemaking), and others have been substantially eroded through interpretive practice (the warrant requirement in Fourth Amendment doctrine, where the catalog of exceptions has grown over time). The grammar's typed-rollback witness might end up in either category.

The Article's response is that the grammar's discipline differs in kind from procedural requirements in familiar legal-doctrinal frameworks. A typed-rollback witness is not a procedure; it is a construction that

either type-checks or does not. The witness cannot be satisfied by going through the motions of construction without producing a valid object. The substrate's runtime kernel evaluates the witness mechanically and rejects ill-typed inputs. A pro forma witness that did not type-check would fail admission at the construction step, not at a later audit step. The discipline is mechanical in a way that traditional procedural requirements are not.

This response has limits. The discipline is mechanical at the level of the substrate. It is not mechanical at the level of the substantive content of the witness. A witness for a file quarantine that specifies a constructive path back to the file's original state is type-checked at the substrate level; whether the constructive path is in fact a meaningful restoration of the prior state is a substantive question that the type system does not resolve. A pro forma witness for a more abstract action – restoring the geopolitical state of a country to its pre-occupation condition, for example – could in principle type-check while failing to provide any meaningful reversibility. The grammar provides a structural discipline at the wrapper level; it does not, by itself, prevent substantive mischief at the content level.

C. The doctrinal-fit objection

A third objection is that the grammar's vocabulary – typed witnesses, type-checking, refinement – is alien to the legal-doctrinal vocabulary in which delegated emergency authority is debated. A legal-academy reader may legitimately wonder whether the structural claim is a genuine addition to the doctrinal discussion or a semantic relabeling of points that the existing doctrinal vocabulary already captures. Sunset clauses, periodic reauthorization requirements, supermajority defaults for continuation, and post-hoc review by independent bodies are all features of the existing doctrinal toolkit. Does the typed-rollback discipline add anything that these existing features do not?

The Article's response is that the typed-rollback discipline adds something specific that the existing toolkit does not directly provide: the requirement is at the admission step rather than at the lapse step or the review step. A sunset clause operates at the lapse step: the authority lapses at a specified time unless extended. The actions taken under the authority are not, by virtue of the sunset, reversed when the authority lapses. A periodic reauthorization requirement operates at a review step: the authority is reviewed periodically and may be continued or terminated. The actions taken between reviews are not, by virtue of the review, reversed. A supermajority default for continuation operates at the extension step: the default is termination unless extended by supermajority. Again, the actions taken under the authority are not reversed by the operation of the default.

The typed-rollback discipline differs from all three. It operates at the admission step. An action that cannot exhibit a constructive path back to the prior state cannot be admitted. The discipline addresses the irreversibility of the action itself, not the lapse, the review, or the continuation of the authority. This is, the Article argues, a genuine addition to the doctrinal toolkit rather than a relabeling of an existing feature.

The argument is not knockdown. A legal-academy reader may reasonably observe that the doctrinal toolkit's mix of sunsets, reauthorizations, and supermajority defaults addresses the irreversibility problem indirectly, by making the authority itself less likely to persist long enough for irreversibility to matter. The typed-rollback discipline addresses the problem directly. Whether the direct address is preferable to the

indirect address is a substantive question on which the Article does not need to take a position; the Article needs only the narrower claim that the direct address is structurally distinct from the indirect ones and is not a relabeling.

D. The cross-disciplinary asymmetry

A final limit is the cross-disciplinary asymmetry of expertise between the Article's home discipline (formal methods, applied to programmable governance) and the Article's target audience (legal academy, particularly constitutional law). The author is not a lawyer. The Article relies on legal historiography, doctrinal accounts, and case-study literatures that a constitutional-law scholar would treat with greater facility than a non-lawyer can. The Article has tried, in each case study and in the doctrinal framing, to defer to the secondary literature where the Article's claim depends on a doctrinal characterization, and to flag uncertainty where the doctrinal record is contested.

The cross-disciplinary asymmetry is, on the author's view, a reason for the Article to be co-authored or extensively reviewed by a constitutional-law scholar before submission. The structural claim does not depend on getting any particular doctrinal claim exactly right; it depends on getting the structural pattern across the five case studies correct, which is a claim that survives reasonable variation in the doctrinal characterizations of any individual case. A reader who finds particular doctrinal claims in Part 3 unconvincing is invited to test the structural claim against the case studies that the reader finds most defensible. The grammar's claim is robust to the failure of any one or two case studies; the claim depends on the pattern across the five.

VIII. Conclusion

The Article has argued that delegated emergency authority across five doctrinally disparate regimes shares a structural feature whose absence explains a recurring failure mode. The feature is the requirement that an authority, at the moment of its exercise, exhibit a typed witness for the path back to the prior state. The feature is not currently a part of any of the five statutory or constitutional regimes examined. Its absence is, on the Article's framing, the structural explanation for the ratcheting pattern documented in the historical record of each regime.

The structural correction the Article proposes is a formal grammar drawn from a Lean-attestable substrate developed in a companion technical paper. The grammar requires four components for any bounded executive action: the substantive action, a time-to-live, a typed rollback witness, and a quorum predicate. The grammar makes the unconstructive case unconstructible: an action that cannot exhibit a typed rollback witness at the moment of admission cannot be admitted. The discipline operates at the construction step, not at a later review step.

The Article has not argued that any particular delegated emergency authority is desirable or undesirable on the merits. It has argued that, structurally, an emergency authority enacted without the four components is a defective grant of authority, and that the defect has substantive consequences that the historical record documents.

Three threads of further work are visible. The first is the engagement of constitutional-law scholarship

in detail. The Article's case studies are written by a non-lawyer working from secondary literature. A constitutional-law scholar would write each case study differently, and the variation would be informative. The second is the engagement of the structural correction with the doctrinal toolkit of sunset clauses, periodic reauthorization, and supermajority defaults. The Article has argued that the typed-rollback discipline is a genuine addition to that toolkit rather than a relabeling, but the comparative analysis of when the direct address is preferable to the indirect addresses is open. The third is the application of the grammar to delegated emergency authority in regimes the Article has not examined. Lincoln's habeas suspensions, Article 16 of the French Fifth Republic, the Italian decree-law tradition, the Indian President's Rule, and the Brazilian state of defense regime are all instances of the pattern. Each would test the grammar's robustness against doctrinal variation the Article has not engaged.

The Article's central contribution is the identification of the pattern, the naming of the structural correction, and the demonstration that the correction is buildable. The substantive content of any application of the grammar is a matter of democratic political determination that the grammar deliberately leaves to the political process. What the grammar offers is a structural skeleton within which that determination is made. The skeleton, on the Article's argument, is what the doctrinal record of delegated emergency authority has been missing.